

M-convexity and the Newton polytope of affine Stanley symmetric polynomials on the 321-avoiding class, without dual k -Schur positivity

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21 September 2026 (PROVE c2)

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Abstract

The substance of this note is a *route*, not a conclusion. Wang, Zhang and Zhang [7] already prove that $\tilde{F}_w(x_1, \dots, x_r)$ is M-convex for *every* affine permutation w , and combining their theorem with Lam’s monomial dominance theorem [4, `thm:monomial`] already identifies its Newton polytope as the $\mu(w)$ -permutahedron for every w . Neither statement is new here. What their proof consumes, and what the motive paragraph preceding their Problems 5.1–5.2 objects to, is Lam’s [5, Cor. 8.5] — dual k -Schur positivity, a deep result.

We give, for *321-avoiding* w , a proof of both statements that does not consume [5, Cor. 8.5], does not consume Lee’s Corollary 5, and mentions dual k -Schur functions nowhere. It consumes instead: the author’s tableau-theoretic theorem on cylindric skew Schur polynomials [2] (which is the answer to Problem 5.2, and which consumes nothing external), and two theorems of *Lam 2006* — a different and elementary paper — namely `thm:321` and `thm:monomial`.

The two missing links are supplied here and were the whole job. Lemma 3.1 (*truncation*) shows that both definitions — the r -factor definition of $\tilde{F}_w(x_1, \dots, x_r)$ and the ℓ -row definition of $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ — are the specialisations $x_{r+1} = x_{r+2} = \dots = 0$ of their own symmetric functions, so an identity in Λ transports coefficientwise to every finite alphabet. Lemma 4.1 (*stabilisation*) shows that the partition $\hat{\lambda}$ produced by the greedy chain of [2] does not depend on the number of rows at all: it is undefined below the minimal chain length and constant above it. This closes gap (iii) of [2] and corrects its statement, which asserted growth.

Two restrictions are stated out loud and not softened. The result covers only the 321-avoiding class. And “without [5, Cor. 8.5]” is the *motive* behind WZZ’s Problem 5.1, not its text: the text asks for a combinatorial proof from the cyclically-decreasing-decomposition definition, and the route here swaps one import for another rather than working from that definition.

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1 Introduction

1.1 What is already known, and by whom

Let $n = k + 1 \geq 2$ and let $\tilde{S}_n$ be the affine symmetric group with generators $s_0, \dots, s_{n-1}$, indices read modulo n .

- **M-convexity of $\tilde{F}_w(x_1, \dots, x_r)$ for all w** is [7, thm-SNP-affine-Ssf]. Their route is: affine Stanley symmetric functions expand positively in dual k -Schur functions [5, Cor. 8.5]; dual k -Schur polynomials are M-convex [7, Thm. 4.4]; a positive sum of M-convex polynomials with a common dominance maximum is M-convex.
- **The Newton polytope for all w .** [7, thm-SNP-affine-Schur-combin] together with [4, thm:monomial] gives $\text{Newton}(\tilde{F}_w) = \mathcal{P}_{\mu(w)}$, where $\mu(w)$ is the conjugate of the decreasing rearrangement of the code $c(w^{-1})$.
- **M-convexity of cylindric skew Schur polynomials** is [7, thm-SNP-cylindric-skew-Schur-function], obtained the same way.

So nothing in the list of conclusions below is new. What is at issue is the input. The paragraph of [7] that introduces Problems 5.1 and 5.2 says, verbatim, that “*it would be interesting to find a direct proof of Theorem [thm-S-M-affine-Ssf] based on the definition given by (def-affine-stanley)*”, and Problem 5.1 then asks for a *combinatorial* proof of the M-convexity of affine Stanley symmetric polynomials, Problem 5.2 for one for cylindric skew Schur polynomials *based on their tableau interpretation*.

1.2 What this note does

[2] answers Problem 5.2: it proves, from the tableau (bead-chain) interpretation and consuming no external theorem at all, that

$$\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_\ell)) = \{\alpha \in \mathbb{N}^\ell : \text{sort}(\alpha) \trianglelefteq \hat{\lambda}\} = \mathcal{P}_{\hat{\lambda}} \cap \mathbb{Z}^\ell, \quad (1)$$

which is M-convex, with $\hat{\lambda}$ computed by an explicit greedy chain.

This note transports (1) across Lam's theorem that a 321-avoiding w has $\tilde{F}_w$ equal to a cylindric skew Schur function. The transport is not automatic: (1) is a statement in ℓ variables, Lam's theorem is an identity in infinitely many, and Problem 5.1 is about r variables. Three alphabets. Section 3 does the bookkeeping, Section 4 removes the dependence of $\hat{\lambda}$ on the alphabet, and Section 6 composes.

1.3 What is honestly new

1. The **route**: a proof of M-convexity and of the Newton polytope for 321-avoiding w that does not consume [5, Cor. 8.5].
2. Lemma 4.1, which **closes and corrects** gap (iii) of [2].
3. Corollary 6.3, which **discharges Hypothesis (H4)** of [3] on the 321-avoiding class — that paper's single remaining gap, there proved for nothing.
4. Corollary 6.4, a combinatorial identity that appears to be new even though both of its sides are old: *the greedy cylindric chain from μ to $u_w \cdot \mu$ has weight sorting to the conjugate of the sorted code of w^{-1} .*
5. A definitional wrinkle in [4] (Remark 5.2), with an explicit witness.
6. Observation 7.2: of the 178 triples (w, r, t) in the range of [3, §9] at which the prefix set A_t has no maximum in the right weak order, **zero** have w 321-avoiding.

2 Definitions, and the three alphabets

2.1 Affine Stanley symmetric polynomials

An element $v \in \tilde{S}_n$ is *cyclically decreasing* if it has a reduced word in which all generators are distinct and s_{i+1} precedes s_i whenever both occur; equivalently $v = u_S$ for a proper subset $S \subsetneq \mathbb{Z}/n\mathbb{Z}$, and $\ell(u_S) = |S|$. The identity is $u_\emptyset$ and is the unique cyclically decreasing element of length 0.

Definition 2.1. For $w \in \tilde{S}_n$ and a finitely supported $\alpha \in \mathbb{N}^{(\infty)}$, let

$$a_\alpha(w) = \#\{(w^1, w^2, \dots) : w = w^1 w^2 \dots, \text{ each } w^t \text{ cyclically decreasing, } \ell(w^t) = \alpha_t, \sum_t \ell(w^t) = \ell(w)\},$$

the product being finite because $w^t = e$ for t beyond the support of α . Put

$$\tilde{F}_w(X) = \sum_{\alpha \in \mathbb{N}^{(\infty)}} a_\alpha(w) x^\alpha \in \Lambda, \quad \tilde{F}_w(x_1, \dots, x_r) = \sum_{\alpha \in \mathbb{N}^r} a_\alpha(w) x^\alpha.$$

This is [4, Alternative Definition 2, l.853–861], and it is verbatim [7, (def-affine-stanley)]. Each monomial of $\tilde{F}_w(X)$ occurs once, because a monomial determines α ; there is no double counting to worry about. The polynomial $\tilde{F}_w(x_1, \dots, x_r)$ is the object of Problem 5.1.

2.2 Cylindric skew Schur polynomials

We use the bead model of [2, §2]. Fix $n > m \geq 1$. A *cylindric shape of type (n, m)* is a strictly increasing $x : \mathbb{Z} \rightarrow \mathbb{Z}$ with $x_{i+m} = x_i + n$; write $\mathcal{C}^{n,m}$ for the set of these and $u(x) = \sum_{i=1}^m x_i$. Write $\nu \prec \rho$ when $\nu_i \leq \rho_i < \nu_{i+1}$ for all i . For $\mu \subseteq \lambda$ in $\mathcal{C}^{n,m}$, $\mathcal{T}_\ell(\lambda/\mu)$ is the set of chains $\mu = x^0 \prec x^1 \prec \dots \prec x^\ell = \lambda$, and such a chain has weight $\alpha_t = u(x^t) - u(x^{t-1})$. Then

$$s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = \sum_{T \in \mathcal{T}_\ell(\lambda/\mu)} x^{\alpha(T)}, \quad s_{\lambda/\mu}^c(X) = \sum_{\ell \geq 0} \sum_{\substack{T \in \mathcal{T}_\ell(\lambda/\mu) \\ \alpha_\ell(T) > 0}} x^{\alpha(T)},$$

the second sum being the usual cylindric skew Schur function; [2, Prop. 2.4] identifies this with the lattice-path definition used by [7].

2.3 Two facts about the number 0

The following two remarks are the entire content of Lemma 3.1, and are separated out because each is the place where the corresponding definition could have failed to be stable under enlarging the alphabet.

Lemma 2.2. $a_{(\alpha,0)}(w) = a_\alpha(w)$ for every α : appending a zero part to a composition does not change the number of decompositions.

Proof. A factorisation counted by $a_{(\alpha,0)}(w)$ has a last factor of length 0; the unique cyclically decreasing element of length 0 is e ; deleting it is a bijection onto the factorisations counted by $a_\alpha(w)$. $\square$

Lemma 2.3. If $\nu \prec \rho$ and $u(\rho) = u(\nu)$ then $\rho = \nu$; and $\nu \prec \nu$ always. Hence inserting a repeat into a chain is a bijection between $\mathcal{T}_\ell$ -chains of weight α and $\mathcal{T}_{\ell+1}$ -chains of weight $(\alpha, 0)$.

Proof. $\nu \prec \rho$ gives $\nu_i \leq \rho_i$ for all i , and $u(\rho) - u(\nu) = \sum_{i=1}^m (\rho_i - \nu_i)$ is a sum of non-negative terms; it vanishes iff every term does. For the second claim, $\nu_i \leq \nu_i < \nu_{i+1}$ holds because ν is strictly increasing. $\square$

3 (D1) Truncation commutes with both definitions

Let $\pi_r : \Lambda \rightarrow \mathbb{Z}[x_1, \dots, x_r]$ be the specialisation $x_j \mapsto 0$ for $j > r$.

Lemma 3.1 (Truncation). For every $w \in \tilde{S}_n$, every cylindric skew shape λ/μ and every $r \geq 1$,

$$\pi_r(\tilde{F}_w(X)) = \tilde{F}_w(x_1, \dots, x_r), \quad \pi_r(s_{\lambda/\mu}^c(X)) = s_{\lambda/\mu}^c(x_1, \dots, x_r).$$

Consequently, if $\tilde{F}_w(X) = s_{\lambda/\mu}^c(X)$ in Λ , then for every $r \geq 1$

$$\tilde{F}_w(x_1, \dots, x_r) = s_{\lambda/\mu}^c(x_1, \dots, x_r)$$

as polynomials — coefficientwise, not merely on supports.

Proof. π_r kills a monomial x^α iff $\alpha_j > 0$ for some $j > r$, and fixes it otherwise. Hence $\pi_r(\tilde{F}_w(X)) = \sum_{\beta \in \mathbb{N}^r} a_{(\beta, 0, 0, \dots)}(w) x^\beta$, and by Lemma 2.2 applied t times, $a_{(\beta, 0, \dots, 0)}(w) = a_\beta(w)$ for every number of appended zeros. This is Definition 2.1 in r variables.

For the cylindric side, the coefficient of x^β ($\beta \in \mathbb{N}^r$) in $s_{\lambda/\mu}^c(x_1, \dots, x_r)$ is $\#\{T \in \mathcal{T}_r : \alpha(T) = \beta\}$, and the coefficient of x^β in $s_{\lambda/\mu}^c(X)$ is $\#\{T \in \mathcal{T}_\ell : \alpha(T) = (\beta, 0, \dots, 0)\}$ for any $\ell \geq r$; by Lemma 2.3 these two counts are equal. The last statement follows because π_r is a ring homomorphism, in particular a well-defined map on Λ . $\square$

Remark 3.2. The brief that set this task predicted that Lemma 3.1 would be “a paragraph”. It is, but the paragraph has content, and the content is Lemmas 2.2 and 2.3: both definitions are sums over combinatorial objects of *unbounded* length, and the truncation statement is exactly the assertion that padding with trivial factors (resp. repeated rows) is a bijection. Had either padding been non-canonical — had there been, say, two cyclically decreasing elements of length 0 — the specialisation would not have matched the r -variable definition and every support statement below would have been about the wrong object.

4 (D2) $\hat{\lambda}$ does not depend on the alphabet

Recall from [2, §5] the *greedy chain* of a cylindric skew shape λ/μ :

$$g^0 = \mu, \quad g_i^t = \min(g_{i+1}^{t-1} - 1, \lambda_i) \quad (i \in \mathbb{Z}, t \geq 1), \quad (2)$$

with $\gamma_t = u(g^t) - u(g^{t-1})$ and $\hat{\lambda}(\lambda/\mu, \ell) = \text{sort}(\gamma_1, \dots, \gamma_\ell)$. Observe at once that (2) *contains no* ℓ . Set

$$\ell_{\min}(\lambda/\mu) = \min\{t \geq 0 : g^t = \lambda\} \in \mathbb{N} \cup \{\infty\}.$$

Lemma 4.1 (Stabilisation). *Let λ/μ be a cylindric skew shape with $\ell_{\min} = \ell_{\min}(\lambda/\mu) < \infty$. Then*

1. $g^t = \lambda$ for every $t \geq \ell_{\min}$, and hence $\gamma_t = 0$ for $t > \ell_{\min}$;
2. $\gamma_t > 0$ for $1 \leq t \leq \ell_{\min}$;
3. $\mathcal{T}_\ell(\lambda/\mu) \neq \emptyset$ if and only if $\ell \geq \ell_{\min}$;
4. for every $\ell \geq \ell_{\min}$, $\hat{\lambda}(\lambda/\mu, \ell) = \text{sort}(\gamma_1, \dots, \gamma_{\ell_{\min}})$. In particular $\hat{\lambda}$ is independent of ℓ on the whole range where it is defined, and it has exactly $\ell_{\min}$ (positive) parts.

If $\ell_{\min} = \infty$ then $\mathcal{T}_\ell(\lambda/\mu) = \emptyset$ for every ℓ and $s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = 0$.

Proof. (1) If $g^{t-1} = \lambda$ then $g_i^t = \min(\lambda_{i+1} - 1, \lambda_i) = \lambda_i$, because λ is strictly increasing gives $\lambda_i \leq \lambda_{i+1} - 1$. Induct.

(2) By [2, Lem. 5.2(1)], $g^{t-1} \ll g^t$ for all $t \geq 1$; in particular $g_i^{t-1} \leq g_i^t$ for all i . If $\gamma_t = 0$ then by Lemma 2.3 $g^t = g^{t-1}$, and then (2) gives $g^s = g^{t-1}$ for all $s \geq t-1$; since $g^{t-1} \neq \lambda$ when $t \leq \ell_{\min}$, this contradicts $\ell_{\min} < \infty$.

(3) ($\Leftarrow$) By (1) and [2, Lem. 5.2(1)], $(g^0, g^1, \dots, g^\ell)$ is a chain in $\mathcal{T}_\ell(\lambda/\mu)$ for $\ell \geq \ell_{\min}$; the steps beyond $\ell_{\min}$ are $\lambda \ll \lambda$, legal by Lemma 2.3. ($\Rightarrow$) Let $T = (x^0, \dots, x^\ell) \in \mathcal{T}_\ell$. By [2, Lem. 5.2(3)], $x_i^t \leq g_i^t$ for all t, i ; at $t = \ell$ this gives $\lambda_i \leq g_i^\ell$, and $g^\ell \subseteq \lambda$ always, so $g^\ell = \lambda$ and $\ell \geq \ell_{\min}$.

(4) Immediate from (1) and (2): $\text{sort}(\gamma_1, \dots, \gamma_\ell)$ discards the zeros $\gamma_{\ell_{\min}+1}, \dots, \gamma_\ell$ and keeps the $\ell_{\min}$ positive parts.

For the last sentence, if some $\mathcal{T}_\ell \neq \emptyset$ then the ($\Rightarrow$) argument of (3) forces $g^\ell = \lambda$, i.e. $\ell_{\min} \leq \ell < \infty$. $\square$

We write $\hat{\lambda}(\lambda/\mu)$ from now on, and $\mathcal{P}_{\hat{\lambda}}$ for the permutahedron it spans (inside $\mathbb{R}^\ell$ for any $\ell \geq \ell_{\min}$).

Remark 4.2 (A correction to [2]). Gap (iii) of [2, §10] reads: “ $\hat{\lambda}(\lambda/\mu, \ell)$ grows with ℓ until ℓ reaches the minimal chain length, after which it is constant. I have not written out the (easy) monotonicity statement.” Lemma 4.1 shows that the first sentence is *false as stated*: there is no growth regime. Below $\ell_{\min}$ the polynomial is identically zero and $\hat{\lambda}$ is not defined; at and above $\ell_{\min}$ it is constant. The gap is closed, and the description of it is withdrawn. I record this because the phrase “easy, not written out” is, in my experience, what usually precedes a correction, and this is one.

5 The bridge: Lam’s theorem on 321-avoiding elements

Definition 5.1. $w \in \tilde{S}_n$ is *321-avoiding* if no reduced word for w contains a braid factor $s_i s_{i+1} s_i$ (equivalently, w is fully commutative); equivalently [4, Prop. in `sec:321`], there are no $x < y < z$ in $\mathbb{Z}$ with $w(x) > w(y) > w(z)$.

Remark 5.2 (Subsequence versus factor). [4, `sec:321`] states the definition with the word “subsequence”: “no reduced word for w contains a subsequence of the form $i(i+1)i$ ”. Read literally that is strictly stronger than the pattern criterion of his own Proposition immediately following. The smallest witness is $w = (-3, 4, 5) \in \tilde{S}_3$ (window notation), of length 4, whose *unique* reduced word is $s_0 s_1 s_2 s_0$: this contains 0, 1, 0 as a subsequence, but no braid factor, and w has no 321 pattern — so the literal reading would call it non-avoiding and the Proposition would be false. The proof of that Proposition (“so that $w = v s_i s_{i+1} s_i u$ ”) shows that the braid *factor* is what is meant. I have checked the two criteria against each other on all 531 elements of $\tilde{S}_n$ with $n \in \{3, 4, 5\}$ and $\ell(w)$ up to 7, 6, 5 respectively: they agree in every case with the factor reading, and disagree on 12 of them with the literal subsequence reading. This is a wording slip, not a mathematical error, but it is worth recording because the two readings are easy to confuse and give different theorems.

We use exactly two theorems of [4], both proved there by elementary manipulations in the affine nilCoxeter algebra $\mathbb{U}_n$, neither depending on [5].

Theorem 5.3 ([4, `thm:321`, 1.1844–47]). *Let $w \in \tilde{S}_n$ be 321-avoiding. Then $\tilde{F}_w$ is equal to a cylindric skew Schur function: there are $N > m \geq 1$ and cylindric shapes $\mu \subseteq \lambda$ in $\mathcal{C}^{N,m}$ with $\tilde{F}_w(X) = s_{\lambda/\mu}^c(X)$.*

Theorem 5.4 (Monomial dominance, [4, `thm:monomial`, 1.945]). *Let $w \in \tilde{S}_n$, let $c'(w) = c(w^{-1})$ be the code of w^{-1} , so $c'_{w(i)} = \#\{j < i : w(j) > w(i)\}$, and let $\mu(w)$ be the conjugate of the decreasing rearrangement of $(c'_1, \dots, c'_n)$. Then $[m_\nu] \tilde{F}_w \neq 0$ implies $\nu \trianglelefteq \mu(w)$, and $[m_{\mu(w)}] \tilde{F}_w = 1$.*

Lemma 5.5. *$\tilde{F}_w(X)$ is monomial-positive, and for $\alpha \in \mathbb{N}^{(\infty)}$ one has $\alpha \in \text{supp } \tilde{F}_w(X)$ if and only if $[m_{\text{sort } \alpha}] \tilde{F}_w \neq 0$. Consequently the set $P(w) = \{\text{sort } \alpha : \alpha \in \text{supp } \tilde{F}_w(X)\}$ has $\mu(w)$ as its unique dominance-maximal element.*

Proof. The coefficients $a_\alpha(w)$ of Definition 2.1 are cardinalities, hence non-negative; $\tilde{F}_w$ is symmetric [4, `thm:sym`, 1.626]; so writing $\tilde{F}_w = \sum_\nu [m_\nu] \tilde{F}_w \cdot m_\nu$ the coefficient of x^α is $[m_{\text{sort } \alpha}] \tilde{F}_w$, and there is no cancellation. The last claim is Theorem 5.4: $\mu(w) \in P(w)$ since $[m_{\mu(w)}] \tilde{F}_w = 1 \neq 0$, and every element of $P(w)$ is $\trianglelefteq \mu(w)$. $\square$

6 Main theorem

Theorem 6.1. *Let $w \in \tilde{S}_n$ be 321-avoiding and let $r \geq 1$. Then*

$$\text{supp}(\tilde{F}_w(x_1, \dots, x_r)) = \{\alpha \in \mathbb{N}^r : \text{sort}(\alpha) \trianglelefteq \mu(w)\} = \mathcal{P}_{\mu(w)} \cap \mathbb{Z}^r,$$

this set is M-convex, and $\tilde{F}_w(x_1, \dots, x_r)$ has saturated Newton polytope with $\text{Newton}(\tilde{F}_w(x_1, \dots, x_r)) = \mathcal{P}_{\mu(w)}$. Moreover $\mu(w) = \hat{\lambda}(\lambda/\mu)$ for any cylindric realisation λ/μ of w as in Theorem 5.3, so $\mu(w)$ is also computed by the greedy chain (2).

The proof consumes Theorems 5.3 and 5.4 of [4], [2, Thm. 7.1], and nothing else. It does not consume [5, Cor. 8.5], Lee's Corollary 5, dual k -Schur functions, k -Kostka positivity, Rado's theorem, or Postnikov's symmetry theorem.

Proof. Fix a realisation $\tilde{F}_w(X) = s_{\lambda/\mu}^c(X)$ from Theorem 5.3, with λ/μ a cylindric skew shape of type (N, m) . Since $\tilde{F}_w \neq 0$ (the identity decomposition of w into $\ell(w)$ factors of length ≤ 1 is counted, or simply: the coefficient of $x_1 x_2 \cdots x_{\ell(w)}$ is the number of reduced words), we have $\ell_{\min} := \ell_{\min}(\lambda/\mu) < \infty$ by Lemma 4.1. Write $\hat{\lambda} = \hat{\lambda}(\lambda/\mu)$, a partition of $\ell(w)$ with exactly $\ell_{\min}$ parts.

Step 1: transport. By Lemma 3.1, $\tilde{F}_w(x_1, \dots, x_r) = s_{\lambda/\mu}^c(x_1, \dots, x_r)$ coefficientwise for every $r \geq 1$; in particular the two supports in $\mathbb{N}^r$ coincide.

Step 2: the support. By [2, Thm. 7.1] applied with $\ell = r$ — legitimate for $r \geq \ell_{\min}$, since that theorem assumes $\mathcal{T}_r(\lambda/\mu) \neq \emptyset$, which is Lemma 4.1(3) — we get $\text{supp}(s_{\lambda/\mu}^c(x_1, \dots, x_r)) = \{\alpha \in \mathbb{N}^r : \text{sort } \alpha \trianglelefteq \hat{\lambda}\}$, M-convex, with Newton polytope $\mathcal{P}_{\hat{\lambda}}$, where by Lemma 4.1(4) the partition $\hat{\lambda}$ does not depend on r .

Step 3: $\hat{\lambda} = \mu(w)$. Take $r = \ell(w)$; this is legitimate for Step 2 because $\hat{\lambda}$ is a partition of $\ell(w)$ with $\ell_{\min}$ parts, whence $\ell_{\min} \leq \ell(w) = r$. Every $\nu \in P(w)$ is a partition of $\ell(w)$, since $\tilde{F}_w$ is homogeneous of degree $\ell(w)$; hence $\ell(\nu) \leq \ell(w) = r$, so ν already occurs as $\text{sort } \alpha$ for some $\alpha \in \mathbb{N}^r$. Therefore Steps 1 and 2 give

$$P(w) = \{\text{sort } \alpha : \alpha \in \text{supp } \tilde{F}_w(x_1, \dots, x_r)\} = \{\nu : \nu \trianglelefteq \hat{\lambda}\}.$$

This set has $\hat{\lambda}$ as its unique dominance maximum. By Lemma 5.5 it has $\mu(w)$ as its unique dominance maximum. A finite poset has at most one maximum, so $\hat{\lambda} = \mu(w)$.

Step 4: small r . It remains to check the displayed identity for $1 \leq r < \ell_{\min}$. There the left-hand side is empty, because $\mathcal{T}_r(\lambda/\mu) = \emptyset$ by Lemma 4.1(3) and Step 1. The right-hand side is empty too. Indeed, suppose $\alpha \in \mathbb{N}^r$ had $\nu := \text{sort } \alpha \trianglelefteq \mu(w) = \hat{\lambda}$. Both are partitions of $d = \ell(w)$, and $\ell(\nu) \leq r < \ell_{\min} = \ell(\hat{\lambda})$. But then $d = \nu_1 + \cdots + \nu_{\ell(\nu)} \leq \hat{\lambda}_1 + \cdots + \hat{\lambda}_{\ell(\nu)} < d$, the last inequality because $\hat{\lambda}$ has strictly more positive parts than $\ell(\nu)$. This is a contradiction, so $\nu \trianglelefteq \hat{\lambda}$ always forces $\ell(\nu) \geq \ell(\hat{\lambda})$.

Finally, $\mathcal{P}_{\mu(w)} \cap \mathbb{Z}^r = \{\alpha : \text{sort } \alpha \trianglelefteq \mu(w)\}$ is Rado's theorem, and is used — as in [2, Thm. 7.1] — only to phrase the answer as a polytope; M-convexity, saturation and the description of the support do not depend on it. $\square$

Remark 6.2 (Why there is no circularity). WZZ's own arrow runs the other way: they pass from affine Stanley symmetric functions to cylindric skew Schur functions via Lee's Corollary 5 (cylindric $\Rightarrow$ there exists a 321-avoiding w). We use Lam's arrow, 321-avoiding $\Rightarrow$ cylindric. These are converse statements, and the cylindric input [2] is proved from the tableau definition with no affine-Stanley input at all, so nothing is assumed that is later derived.

Corollary 6.3 (Discharge of (H4) on the 321-avoiding class). *Let $w \in \tilde{S}_n$ be 321-avoiding and $r \geq 1$ with $\tilde{F}_w(x_1, \dots, x_r) \neq 0$. Then there is a cyclically decreasing decomposition $w = w^1 w^2 \dots w^r$ that maximises $\ell(w^1 \dots w^t)$ for every $t = 1, \dots, r$ simultaneously. This is Hypothesis (H4) of [3, §5], which is there assumed and not proved.*

Proof. Take the decomposition with $\alpha = \mu(w)$ padded with zeros, which lies in the support by Theorem 6.1. For any β in the support, $\text{sort } \beta \leq \mu(w)$ gives $\beta_1 + \dots + \beta_t \leq (\text{sort } \beta)_1 + \dots + (\text{sort } \beta)_t \leq \mu(w)_1 + \dots + \mu(w)_t$ for every t . Since $\ell(w^1 \dots w^t) = \alpha_1 + \dots + \alpha_t$ by length-additivity, the decomposition realising $\alpha = \mu(w)$ attains the maximum at every t at once. $\square$

Corollary 6.4 (Greedy chain computes the code). *Let $w \in \tilde{S}_n$ be 321-avoiding and let $u_w \cdot \mu = \lambda$ be a realisation in $\mathcal{C}^{N,m}$ as in [4, §7]. Then the weight γ of the greedy chain (2) of λ/μ satisfies*

$$\text{sort}(\gamma) = \mu(w) = (\text{decreasing rearrangement of } c(w^{-1}))',$$

and the minimal number of rows of a cylindric tableau of shape λ/μ equals the number of parts of $\mu(w)$, i.e. the largest entry of $c(w^{-1})$.

Proof. Step 3 of Theorem 6.1, plus Lemma 4.1(4) for the count of parts. $\square$

7 Verification

7.1 What may and may not be tested

M-convexity of $\tilde{F}_w(x_1, \dots, x_r)$ is a published theorem of [7], valid for all w . *Every consequence of it is therefore unfalsifiable by computation:* a run that confirms M-convexity tests my code, not my claim. I ran such a check anyway as a code control — 1038 pairs (w, r) with non-empty support over $n \leq 5$, 0 failures — and it certifies only that the implementation of Definition 2.1 is not broken.

The observables that can genuinely fail are properties of the *route*: the dictionary between the two models, the stabilisation of $\hat{\lambda}$, and the identity $\hat{\lambda} = \mu(w)$. Those are reported below.

7.2 The instrument

To test the route I implemented the `nilCoxeter` action u_i of [4, §7] in the bead coordinates of [2]. The convention was *derived, not fitted*. In the bead model a shape has bead set $A \subseteq \mathbb{Z}$ with $A + N = A$; adding one box per period to row j means $x_j \mapsto x_j + 1$, legal exactly when $x_j + 1 \notin A$, i.e. it turns the pattern $(q_a, q_{a+1}) = (1, 0)$ into $(0, 1)$ where q is the indicator of A . Lam states his rule on the edge sequence p as $(p_i, p_{i+1}) = (0, 1) \mapsto (1, 0)$. Matching the two forces $p = 1 - q$ (his p is the *hole* indicator) and forces his diagonal index to be my bead position, read in the same increasing direction. The only remaining freedom is the origin of the labelling, and $\tilde{F}_w$ is invariant under rotation of the labels [4, Prop. at l.1113]. So:

$$u_i \cdot x = \begin{cases} \text{move every bead at a position } \equiv i \pmod{N} \text{ one step right,} & i \bmod N \in \bar{A}, (i+1) \bmod N \notin \bar{A}, \\ 0 & \text{otherwise,} \end{cases}$$

where $\bar{A} = A/N\mathbb{Z}$.

7.3 Route observables

All ranges: $n = 3, \dots, 6$; $\ell(w)$ up to 9, 7, 6, 5 respectively; r up to 6, 5, 4, 4. Code at `proofs/code-q215/` (`cylact.py`, `verify_q215.py`, `q212.py`); `verify_q215.py` reproduces every number in this subsection in one run.

- T0 u_i is a representation.** $u_w \cdot \mu$ computed along *every* reduced word of w : 7886 reduced words, all agreeing. (Could fail: a transcription error in the action would generically break reduced-word independence.)
- T1 Theorem 5.3 and its converse.** 637 321-avoiding w : every one admits a cylindric shape μ with $u_w \cdot \mu \neq 0$. 708 non-321-avoiding w : *none* does. 0 failures either way. (Could fail: it is a sharp dichotomy and a wrong 321 test or a wrong action would break it.)
- T2 The dictionary, coefficientwise (Lemma 3.1).** $\tilde{F}_w(x_1, \dots, x_r)$ computed from Definition 2.1 by dynamic programming over the affine weak order, versus $s_{\lambda/\mu}^c(x_1, \dots, x_r)$ computed from bead chains: 4794 instances, agreement in every coefficient, 0 failures. **This is the test that can kill the route**, and in particular it is the test that would have exposed a reversed orientation in the bead labelling. The two sides share no code.
- T3 Stabilisation (Lemma 4.1).** 4568 checks of $\hat{\lambda}(\lambda/\mu, \ell) = \hat{\lambda}(\lambda/\mu, \ell_{\min})$ for $\ell_{\min} \leq \ell \leq \ell_{\min} + 3$: 0 failures.
- T4 $\hat{\lambda} = \mu(w)$ (Corollary 6.4).** 1142 instances: the greedy bead chain and the conjugate of the sorted code of w^{-1} — two routes sharing no code and no tuned parameter — agree in every case. 0 failures.

Total: 19,735 route checks, 0 failures.

Remark 7.1. T4 is a theorem (Corollary 6.4), so its agreement is not independent evidence *for the mathematics*. It is independent evidence for the *dictionary*: a mismatch would have been fatal to the bead implementation of u_w or to the greedy recursion, and it is precisely the sort of error that a shape match alone would not reveal.

7.4 Observation: where the weak-order route works

Observation 7.2. Section 9 of [3] records that the natural route to (H4) — that the set A_t of achievable prefixes $w^1 \cdots w^t$ has a maximum in the right weak order — is false: over $n \in \{3, 4\}$, $\ell(w) \leq 5$, $r \leq 4$ there are 767 triples (w, r, t) with $|A_t| > 1$, and in 178 of them A_t has no maximum. I reimplemented that computation independently and reproduced both numbers, and then split by 321-avoidance:

	all w	w 321-avoiding
triples (w, r, t) with $ A_t > 1$	767	410
of these, A_t has no right-weak maximum	178	0

Every failure of the weak-order route has w *not* 321-avoiding.

This is consistent with, and suggested by, Theorem 6.1: on the 321-avoiding class the greedy chain (2) supplies a canonical simultaneous maximiser (this is Corollary 6.3), whereas off that class there is no cylindric shape to run it on. I do *not* claim the two statements are equivalent: Corollary 6.3 is about maxima of *lengths*, Observation 7.2 about maxima in the *weak order*, and the second is strictly stronger. Whether 321-avoidance characterises the w for which every A_t has a weak-order maximum is open, and is the first question I would ask next.

8 What this proof consumes, and what I did not verify

8.1 External theorems used

- [4, `thm:321`]: 321-avoiding $\Rightarrow$ cylindric. *Used*, Theorem 6.1 Step 1. Read at source (arXiv long version).
- [4, `thm:monomial`]: monomial dominance. *Used*, Lemma 5.5 and Step 3 — but only to *name* $\tilde{\lambda}$; M-convexity, saturation and the polytope $\mathcal{P}_{\tilde{\lambda}}$ do not need it.
- [4, `thm:sym`]: $\tilde{F}_w$ is symmetric. *Used*, Lemma 5.5. Read at source; it follows from [4, `prop:commute`], an explicit bijection in $\mathbb{U}_n$.
- [2, Thm. 7.1]: the cylindric support theorem. *Used*, Step 2. By the present author; it consumes nothing external.
- [5, Cor. 8.5], Lee 2019 Cor. 5, dual k -Schur functions, k -Kostka positivity, cores, k -bounded partitions, Postnikov’s symmetry theorem, Murota’s theory beyond the definition of M-convexity: *not used*.
- Rado’s theorem: used only to restate the answer as $\mathcal{P}_{\mu(w)} \cap \mathbb{Z}^r$; not used for any of the mathematical conclusions.

8.2 Structural hypotheses — the things with no citation

A list of citations is not a list of hypotheses. The following are load-bearing and have no reference attached to them, so they are easy to miss.

Structural hypothesis 8.1 (Both definitions are stable under padding). Lemmas 2.2 and 2.3. Proved here. If either failed, Lemma 3.1 would fail and every statement in this note would be about a different polynomial.

Structural hypothesis 8.2 (The two papers’ $s_{\lambda/\mu}^c$ are the same object). Lam’s cylindric Schur function [4, §7], WZZ’s cylindric skew Schur polynomial [7, §5], and the bead-chain generating function of [2, §2] must all agree. *Not proved here*. [2, Prop. 2.4] proves the second = the third. The first = the second is asserted by Lam (“one can check directly using the definition of cylindric semistandard tableaux”, 1.1713–21) and is taken on his authority. Evidence: observable T2 above, 4794 coefficientwise agreements; and the untuned control in [2, §9], in which the bead model reproduces WZZ’s published Example 4.2 — itself an affine Stanley symmetric function — exactly, negative Schur coefficient included.

Structural hypothesis 8.3 ($\ell_{\min} < \infty$ for a realisation). Used in Theorem 6.1. Proved there from $\tilde{F}_w \neq 0$ and Lemma 4.1.

8.3 What I did not verify

1. **A sentence in Lam’s proof of `thm:321` that I cannot follow.** In the branch where the letter i does not occur in the chosen reduced word, the base shape changes from μ to a newly constructed λ , and the text then asserts that “it is clear that $u_w \cdot \lambda \neq 0$ ” while the inductive hypothesis was about μ (1.1868). **I have confirmed that the sentence says this; I have not confirmed that it is a gap.** It may well be clear and I may be

missing the argument. Mitigating: the conclusion is independently supported by observable T1 above (637/637 realisations found, 708/708 correctly refused), and by Lee’s reproof [6] — though Lee’s Corollary 5 is the *converse* direction and his only bijection is restricted to straight shapes and 0-Grassmannian w , so it does not by itself repair this direction.

2. **Numbering.** All references to [4] are by L^AT_EX label (`thm:321`, `thm:monomial`, `thm:sym`, `prop:commute`, `sec:321`) and line number in the arXiv long version. The published Amer. J. Math. version may number differently.
3. **The explicit shape.** Theorem 6.1 uses only the *existence* of a cylindric realisation. The realisations used in §7 were found by my own implementation of u_w , not by transcribing Lam’s inductive construction; so the note does not certify that Lam’s construction outputs the shapes mine does. For the theorem this is irrelevant — any realisation gives the same $\hat{\lambda} = \mu(w)$ by Corollary 6.4 — but it would matter to anyone wanting the shape in closed form.

9 Scope, and what is *not* claimed

1. **This is not a solution of Problem 5.1.** Problem 5.1 asks for a combinatorial proof of M -convexity of affine Stanley symmetric polynomials; its preamble asks for a proof *based on* (`def-affine-stanley`). The route here is not from the definition: it replaces the import [5, Cor. 8.5] with the imports [4, `thm:321`, `thm:monomial`]. One import for another. What can honestly be said is: *on the 321-avoiding class, M -convexity and the Newton polytope follow without dual k -Schur positivity.*
2. **It covers only the 321-avoiding class,** which is exactly the class on which $\tilde{F}_w$ is a cylindric skew Schur function [4, `sec:321`]. For $n = 3$ and $\ell(w) \leq 7$ that is 39 of 84 elements; for $n = 6$, $\ell(w) \leq 5$, it is a minority.
3. **The M -convexity conclusion itself is not new** — see §1.1. Neither is the identification of the Newton polytope. The novelty is the route, the discharge of (H4) on this class (Corollary 6.3), the closing of gap (iii) of [2] (Lemma 4.1), and Corollary 6.4.
4. **The cylindric half is the stronger half.** [2] answers Problem 5.2 outright and strengthens it, consuming nothing; the present note is a corollary of it. A reader with time for one of the two should read that one.
5. **“Saturation” is overloaded in this literature.** Alexandersson and Kantarcı Oğuz [1] prove “the saturation property for cylindric Schur functions”, meaning Knutson–Tao saturation of cylindric Kostka coefficients ($|CSSYT(kD, k\nu)| > 0 \iff |CSSYT(D, \nu)| > 0$). That is a different statement from saturated Newton polytope: the words *M -convex*, *Newton* and *permutahedron* do not occur in their paper, and their polytopes are Gelfand–Tsetlin polytopes. Nothing here restates or uses their result.

References

- [1] P. Alexandersson and E. Kantarcı Oğuz. *Cylindric Schur functions and the saturation property*. arXiv:2311.07382.

- [2] Clio. *A combinatorial proof of the M-convexity of cylindric skew Schur polynomials*. `projects/proofs/2026-09-20-c1-cylindric-M-convexity.tex`, 20 September 2026; `clio-vega/proofs@68184eb`.
- [3] Clio. *An exchange move for cyclically decreasing factorisations, and the reduction of WZZ Problem 5.1 to a single hypothesis*. `projects/proofs/2026-09-21-c1-affine-stanley-exchange.tex`, 21 September 2026; `clio-vega/proofs@22d1b22`.
- [4] T. Lam. *Affine Stanley symmetric functions*. Amer. J. Math. **128**(6), 1553–1586 (2006); arXiv:math/0501335. (Read at LaTeX source, long version: `def:affineStanley` l.601, Alternative Definition 2 l.853–861, `thm:sym` l.626, `prop:commute` l.655, `prop:Vaffine` l.880, `thm:monomial` l.945, the $\mathbb{U}_n$ -action on cylindric shapes l.1700–21, `sec:321` l.1793, `thm:321` l.1844–47.)
- [5] T. Lam. *Schubert polynomials for the affine Grassmannian*. J. Amer. Math. Soc. **21**, 259–281 (2008); arXiv:math/0603125. (Cited only to say it is *not* used.)
- [6] S. J. Lee. *Cylindric skew Schur functions and affine Stanley symmetric functions*. J. Combin. Theory Ser. A **168**, 26–49 (2019); arXiv:1706.04460. (Corollary 5 at l.378 of the arXiv source; agent-read, not read at first hand by the author.)
- [7] B. Wang, C. X. T. Zhang and Z.-X. Zhang. *Newton polytopes and M-convexity of dual k-Schur polynomials*. arXiv:2401.14632. (Read at LaTeX source: M-convexity definition l.122, `thm-SNP-affine-Ssf` l.875, `thm-SNP-cylindric-skew-Schur-function` l.922–26, Problems 5.1 and 5.2 at l.929–948.)